

● PRO PLAN · \$19.99/MONTH

APPLIED SCIENCE

SUPPLEMENT
GUIDE

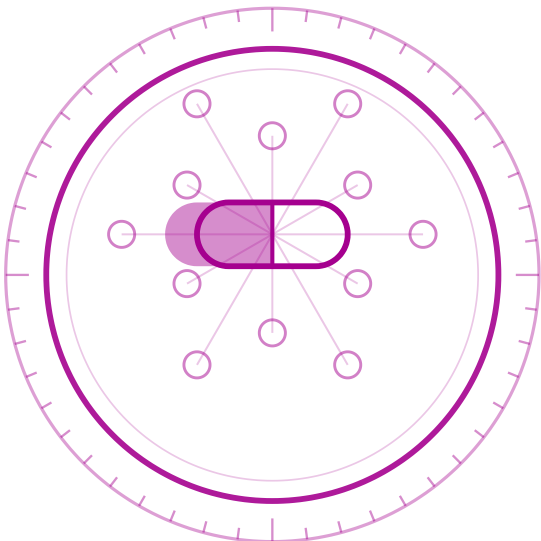
What to take, how much, when and why
— and above all, **what never to buy**.

5 ESSENTIALS

4 OPTIONAL

10 THAT DO NOT WORK

REAL BUDGET



SUPPLEMENTS
Dose and timing

9 EVIDENCE-BACKED	10 MONEY WASTED	~\$55 MONTHLY COST	8 PAGES
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BEFORE YOU SPEND A SINGLE EURO

Supplements contribute, at most, **5 % of the result**. If the other 95 % is not in place, no tub is going to fix it. This is the real hierarchy, in order of impact.

01 Calorie balance and protein

They determine whether you gain or lose, and how much of it is muscle.

02 Training with progressive overload

The stimulus without which there is no adaptation.

03 7-9 hours of sleep

Where recovery and protein synthesis actually happen.

04 Consistency over months

The factor that most separates those who progress from those who do not.

05 Supplementation

The fine tuning. It only makes sense once the above already works.

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THE QUESTION TO ASK YOURSELF IN FRONT OF ANY TUB

Is there at least one meta-analysis backing its effect in healthy humans, at the dose the product actually contains? If the answer is no, you are paying for marketing.

THE TWO ESSENTIALS

1 · PROTEIN POWDER (WHEY OR PLANT-BASED)

It is not magic: 1 g of whey protein builds exactly the same as 1 g of protein from chicken. Its value is **logistical** — it lets you hit your daily quota when real food does not fit your schedule.

PARAMETER	DETAIL
Dose	20–40 g per serving (1–2 scoops)
When	Post-workout or between meals, as a top-up to close the daily total
Type	Concentrate (cheaper) or isolate (less lactose and fat)
Vegan option	Pea + rice in a 70/30 ratio, to complete the amino acid profile
Verified brands	Optimum Nutrition, MyProtein, Garden of Life (vegan)
Indicative price	\$30–60 for 2 kg (1 to 2 months of use)

2 · CREATINE MONOHYDRATE

The most studied supplement in the history of sport, with hundreds of trials and an excellent safety profile in healthy people. It raises strength by 5 to 15 %, speeds recovery between sets and has documented cognitive benefits.

PARAMETER	DETAIL
Dose	3–5 g a day. The loading phase is optional and does not improve the final result
When	Any time of day. Taking it with a meal slightly improves absorption
Type	Monohydrate only. The other forms are more expensive marketing
Cycling	Not needed. Continuous use is safe and more effective
Brands	Creapure (Germany) or any unflavoured monohydrate from a known brand
Indicative price	\$15–25 for 500 g (3–4 months of use)

TWO CLARIFICATIONS ABOUT CREATINE

The 1–2 kg gain of the first weeks is **intramuscular water**, not fat: it is the intended effect. And no, it does not damage the kidneys in healthy people — it can slightly raise creatinine in a blood test, which is an artefact of the measurement, not harm. Tell your doctor you take it before any blood work.

THE THREE FOR HEALTH AND RECOVERY

3 · OMEGA-3 (EPA + DHA)

PARAMETER	DETAIL
Dose	2–3 g of EPA + DHA combined per day (check the back: that is not the same as grams of oil)
Benefits	It lowers inflammation, improves recovery and protects cardiovascular health
When	With meals containing fat, to maximise absorption
Type	Fish oil (salmon, sardine, anchovy) or algae oil for the vegan option
Brands	Nordic Naturals, Solgar, OmegaVia
Indicative price	\$20–35 for 90 capsules

4 · VITAMIN D3 + K2

A large share of the population has insufficient levels, especially with indoor work or in cloudy climates. It is involved in immune function, bone health, recovery and hormone production.

PARAMETER	DETAIL
D3 dose	2,000–4,000 IU daily
K2 dose	100–200 mcg (it directs calcium to bone rather than to the artery)
When	In the morning, with breakfast: it is fat-soluble and needs fat
Form	D3 (cholecalciferol), never D2 — it is absorbed far worse
Monitoring	Ask for 25-OH-vitamin D in your annual blood test and adjust with your doctor
Indicative price	\$10–20 for 3 months

5 · MAGNESIUM

PARAMETER	DETAIL
Dose	300–400 mg of elemental magnesium a day
Benefits	It improves sleep quality and neuromuscular function, and helps modulate cortisol
When	30–60 minutes before sleep
Form	Bisglycinate (better tolerated) or citrate. Avoid the oxide: it is barely absorbed
Indicative price	\$15–25 for 3 months

BUYING ORDER IF YOUR BUDGET IS LIMITED

Creatine → protein → vitamin D3 → magnesium → omega-3. The first two have the greatest impact per euro spent in an athlete who already trains and eats well.

SECOND TIER: PERFORMANCE

They have solid evidence, but their effect is smaller and more specific. Add them only once the previous five are already part of your routine.

SUPPLEMENT	DOSE	WHEN	WHAT TO REALISTICALLY EXPECT
Caffeine	3–6 mg/kg	45–60 min before	More strength, more reps and a lower perception of effort
Beta-alanine	3–5 g/day	Any time, daily	It improves efforts of 1 to 4 minutes. It needs 4 weeks to saturate
Citrulline malate	6–8 g	60 min before	A few more reps and a better muscle pump
Ashwagandha KSM-66	300–600 mg	Night	It helps with perceived stress. Moderate evidence, not spectacular
Electrolytes	According to sweat rate	Long sessions or heat	It prevents cramps and slumps when you sweat a lot

CAFFEINE: HOW TO USE IT WELL

WHAT TO DO

- Start at 3 mg/kg and go up only if you need to
- Combine it with L-theanine in a 1:2 ratio if it makes you jittery
- Save it for your 2–3 hardest sessions of the week
- Take a week off every 6–8 to restore sensitivity

WHAT RUINS THE RESULT

- Taking it after 14:00: its half-life is 5–6 hours
- Using it daily at high doses: tolerance cancels the effect
- Using caffeine to paper over insufficient sleep
- Stacking pre-workout, coffee and an energy drink without counting the total

SAFETY LIMIT

Do not exceed 400 mg of caffeine a day (about 4 coffees), nor 200 mg in a single dose if you have no tolerance. If you have hypertension, arrhythmias, anxiety or sleep problems, check with your doctor before using it as an ergogenic aid.

A note on beta-alanine: the tingling in face and hands (paraesthesia) is harmless and goes away by splitting the dose in two or using the sustained-release version.

THE 10 THAT DO NOT WORK

None of these products is dangerous in itself; they simply do not do what the label promises. This is the money most people throw away every month.

PRODUCT	WHAT IT PROMISES	WHAT THE EVIDENCE SAYS
BCAAs / branched-chain amino acids	More muscle and less catabolism	Useless if you already get enough protein: the other essential amino acids are missing
Thermogenic fat burners	Speeding up metabolism	The effect on expenditure is minimal and does not change body composition
Glutamine	Recovery and hypertrophy	No effect in healthy, well-fed athletes
Testosterone boosters	Raising testosterone	Tribulus and the like do not raise it in men with normal levels
«Advanced» creatines	Better absorption than monohydrate	HCL, ethyl ester and kre-alkalyn: more expensive, with no proven advantage
Detoxes and cleanses	Removing toxins	Your liver and kidneys already handle that, free of charge
Nitric oxide capsules	Pump and vasodilation	They usually contain doses far below the effective amount of citrulline
Megadose multivitamins	Covering every deficiency	On a varied diet you only produce expensive urine; some vitamins accumulate
Collagen for muscle	Gaining muscle mass	A low biological value protein. Useful for tendon, not for hypertrophy
Exogenous ketones	Burning fat and giving energy	Expensive, unpleasant tasting and of no benefit to an athlete who eats carbohydrates

THE 30-DAY RULE

Before buying a new supplement, work out what it costs per year and ask yourself what else you would do with that money: a few more months of creatine and protein? Better-quality real food? Almost always the right answer is the second.

YOUR DAILY PROTOCOL

SUPPLEMENT	MORNING	PRE-WORKOUT	POST-WORKOUT	NIGHT
Protein powder	—	—	CORE	If the quota is short
Creatine 3-5 g	OPTION	Option	WITH THE PROTEIN	Option
Omega-3	WITH BREAKFAST	—	—	Option
Vitamin D3 + K2	REQUIRED	—	—	—
Magnesium	—	—	—	30 MIN BEFORE SLEEP
Caffeine (optional)	—	45-60 MIN BEFORE	—	Never
Beta-alanine (optional)	Option	Option	Option	Option

A REALISTIC MONTHLY BUDGET

LEVEL	INCLUDES	MONTHLY COST
Bare minimum	Creatine + vitamin D3	≈ \$10
Recommended	Creatine + protein + D3 + magnesium	≈ \$40
Full	The 5 essentials + caffeine	≈ \$55
Performance	Complete + beta-alanine + citrulline	≈ \$75

YOUR INVENTORY

SUPPLEMENT	BRAND	DAILY DOSE	RESTOCK DATE
Protein			
Creatine			
Omega-3			
Vitamin D3 + K2			
Magnesium			

WHERE EVERY NUMBER COMES FROM

Every number in this document comes from sources that are public, free and checkable. Not one figure is invented, rounded for convenience or copied from another popular publication. This page exists so that you can verify it yourself.

WHY THESE SOURCES AND NO OTHERS

Works of the United States federal government are not subject to copyright (17 U.S.C. § 105). That makes a handful of agencies the only source of nutrition and training data that is both first-rate and legally free. This is not restricted material: it is public material that almost nobody opens.

WHAT YOU ARE LOOKING FOR	WHAT YOU WILL FIND	WHERE
Reference intakes	RDA, AI and tolerable upper intake level for every vitamin and mineral	ods.od.nih.gov
Food composition	The full profile of any food, nutrient by nutrient	fdc.nal.usda.gov
Dietary guidelines	Dietary patterns and limits for added sugars, saturated fat and sodium	health.gov
Physical activity guidelines	The weekly dose, its response curve and the strength requirement	health.gov
Physical readiness doctrine	FM 7-22: training, nutrition, sleep and mental readiness	armypubs.army.mil
Lifting equation	How much weight is reasonable to lift in each specific situation	cdc.gov/niosh
Food safety	Safe internal temperatures and the danger zone	foodsafety.gov
Health claims	What may legally be claimed about a nutrient in Europe, and what was refused	European Commission register

HOW THIS DOCUMENT CHECKS ITSELF

The reference figures are not typed by hand: they live in a data module that the typesetting program verifies before writing a single line. It checks that the macronutrient ranges can add up to one hundred per cent of energy, that none of the 19 micronutrients has a recommended value above its own ceiling without a note explaining why, and that no safe cooking temperature falls inside the danger zone. If a single check fails, the document is not typeset.

REFERENCE VALUES ARE POPULATION VALUES

They describe what covers the needs of most healthy adults. They are not an individual prescription, they do not account for disease, pregnancy, breastfeeding or medication, and no publication replaces assessment by a registered health professional. If you take medication, some of the nutrients named here interact with it in clinically relevant ways: do not change an intake or start a supplement without telling whoever prescribes for you.

QUALITY, SAFETY AND ANTI-DOPING

HOW TO CHOOSE A RELIABLE PRODUCT

- Third-party testing marks: **Informed Sport** or **NSF Certified for Sport**
- A label with the dose per serving, not a «proprietary blend» with no amounts
- A single active ingredient per tub whenever possible
- A manufacturer with batch, expiry date and country of origin visible

WARNING SIGNS

- Promises of results «in 7 days» or comparisons with drugs
- Proprietary formulas that hide the real doses
- Weight-loss products bought outside regulated channels
- Ingredients you can neither pronounce nor find in the literature

SITUATIONS THAT REQUIRE CHECKING WITH A HEALTH PROFESSIONAL FIRST

- ☐ You take long-term medication: anticoagulants, antihypertensives, antidiabetics, thyroid drugs or psychotropics
- ☐ You have renal, hepatic or cardiac disease, or an eating disorder
- ☐ You are pregnant, breastfeeding or a minor
- ☐ You compete under anti-doping control: check every product against the WADA list
- ☐ You are about to have blood work: mention that you take creatine

FOOD FIRST, THE TUB SECOND

No supplement in this guide makes up for sleeping badly or eating too little: **they are the final 5 %**, not the shortcut. Start with the two essentials, give them eight weeks, and add nothing else until you can measure what changed.

BASIC

\$9.99

Fundamentals
and quick guides

PRO

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Your current plan
Advanced nutrition

ELITE

\$34.99

Everything included
+ one-to-one coaching

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IMPORTANT NOTICE

This guide is for informational and educational purposes only. It does not constitute medical advice, diagnosis, treatment or prescription, and it does not replace assessment by a registered health professional. Food supplements are not intended to prevent, treat or cure any disease, and must not replace a varied, balanced diet. The doses given are for healthy adults; consult your doctor or pharmacist before starting any supplementation, especially if you take medication, have a medical condition, are pregnant or breastfeeding, or are a minor. The brands named are mentioned for guidance only and imply no sponsorship or commercial relationship. Prices are indicative and may vary by country and retailer. Copyrighted content: reproduction, distribution or resale without written permission from VILLUMINATIONS is prohibited.